TRAINING CHESS BOTS WITH VARIOUS MACHINE LEARNING TECHNIQUES

by

BHAVYA SINGH
URN: 6839571

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Department of Computer Science
University of Surrey
Guildford GU2 7XH

Supervised by: Dr Joey Sik Chun Lam

I declare that this dissertation is my own work and that the work of others is acknowledged and indicated by explicit references.

Bhavya Singh
May 2026

Abstract

Write a summary of the work presented in your dissertation. Introduce the topic and highlight your main contributions and results. The abstract should be comprehensible on its own, and should no contain any references. As far as possible, limit the use of jargon and abbreviations, to make the abstract readable by non-specialists in your area. Do not exceed 300 words.

Acknowledgements

Write any personal words of thanks here. Typically, this space is used to thank your supervisor for their guidance, as well as anyone else who has supported the completion of this dissertation, for example by discussing results and their interpretation or reviewing write ups. It is also usual to acknowledge any financial support received in relation to this work.

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Glossary

A_f	The source message, being a sequence of f source symbols $a_1 a_2 \ldots a_f$
a_i	The i^{th} symbol in the source message, where $a_i \in S_m$
B_g	The decoded message, being a sequence of g source symbols $b_1 b_2 \ldots b_g$
b_i	The i^{th} symbol in the decoded message, where $b_i \in S_m$
C_h	The transmitted (compressed) message, being a sequence of h Tunstall codewords $c_1 c_2 \ldots c_h$
c_i	The i^{th} codeword in the transmitted message, where $c_i \in T_n$
D_h	The received (compressed) message, which for a complete Tunstall code is a sequence of h Tunstall codewords $d_1 d_2 \ldots d_h$
d_i	The i^{th} codeword in the received message, where $d_i \in T_n$ for a complete Tunstall code

Abbreviations

BER	Bit Error Rate
BPSK	Binary Phase Shift Keying
BSC	Binary Symmetric Channel
DCT	Discrete Cosine Transform
ECC	Error Correcting Codes
FEC	Forward Error Correction
JPEG	Joint Photographic Experts Group
MPEG	Moving Pictures Experts Group
SER	Symbol Error Rate
SNR	Signal to Noise Ratio

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